Project Report: *ISL Gestures Recognition System*

1. Introduction

Indian Sign Language (ISL) is an important mode of communication for people who are deaf or hard of hearing. However, many people are not familiar with sign language, which creates a communication gap in daily interactions.

This project focuses on building a real-time system that can recognize ISL gestures using a webcam. The aim is to make communication more accessible and demonstrate how machine learning can be used in simple, practical applications.

2. Problem Statement

Communication between sign language users and non-signers can be challenging. Since most people do not understand ISL, even basic interactions like conveying numbers can become difficult.

The problem addressed in this project is:

How can we create a simple system that recognizes ISL gestures in real time and displays them in an understandable format?

3. Objective

The main objectives of this project are:

- To recognize hand gestures representing numbers and some basic words in ISL.
- To provide real-time output using a webcam.
- To make the system simple and user-friendly.
- To explore the use of machine learning in web-based applications.

4. Approach and Methodology

To solve the problem, a machine learning-based approach was used.

First, a model was trained using Google Teachable Machine by collecting multiple samples of hand gestures for different numbers. These samples were used to teach the model to distinguish between different signs.

After training, the model was exported and integrated into a web application using TensorFlow.js. The frontend of the application was built using HTML, CSS, and JavaScript.

The system captures live video through the webcam, processes each frame, and predicts the gesture being shown. To improve accuracy, a stability mechanism was added so that the output is only displayed when the prediction remains consistent for a short period.

5. Key Features

- Real-time gesture detection using webcam.
- Stable output to avoid flickering predictions.
- Voice output for better accessibility.
- Simple interface with start and stop controls.

6. Key Decisions

During the development of this project, several important decisions were made:

- **Using Teachable Machine:** It simplified the model training process without requiring complex coding.
- **Web-based implementation:** Using TensorFlow.js allowed the model to run directly in the browser without installation.
- **Adding stability logic:** Instead of showing every prediction, the system waits for consistent results to improve user experience.

- **Minimal UI design:** The interface was kept clean and simple to focus on functionality.

7. Challenges Faced

Some challenges were encountered during the project:

- The model initially produced unstable and flickering outputs.
- Background noise and lighting conditions affected accuracy.
- Managing real-time predictions efficiently in the browser.
- Integrating voice output with detection logic.

These challenges were addressed by improving the dataset, adding confidence thresholds, and implementing stability checks.

8. Limitations

- The model may sometimes give incorrect predictions.
- Performance depends on lighting and background conditions.
- The dataset used for training is limited.
- Only number gestures are supported, not full sign language.

9. Future Improvements

The project can be improved further by:

- Increasing the size and diversity of the training dataset.
- Adding support for alphabets and more words.
- Improving accuracy with advanced models.
- Enhancing the user interface.
- Adding mobile compatibility.

10. Conclusion

This project successfully demonstrates how machine learning can be applied to recognize ISL gestures in real time. It highlights the potential of AI in improving accessibility and bridging communication gaps.

Although the system has some limitations, it provides a strong foundation that can be expanded into a more advanced and practical solution in the future.

11. What I Learned

Through this project, I learned:

- Basics of machine learning model training.
- How to use Teachable Machine.
- Integration of ML models using TensorFlow.js.
- Frontend development using HTML, CSS, and JavaScript.
- Problem-solving and debugging real-time applications.